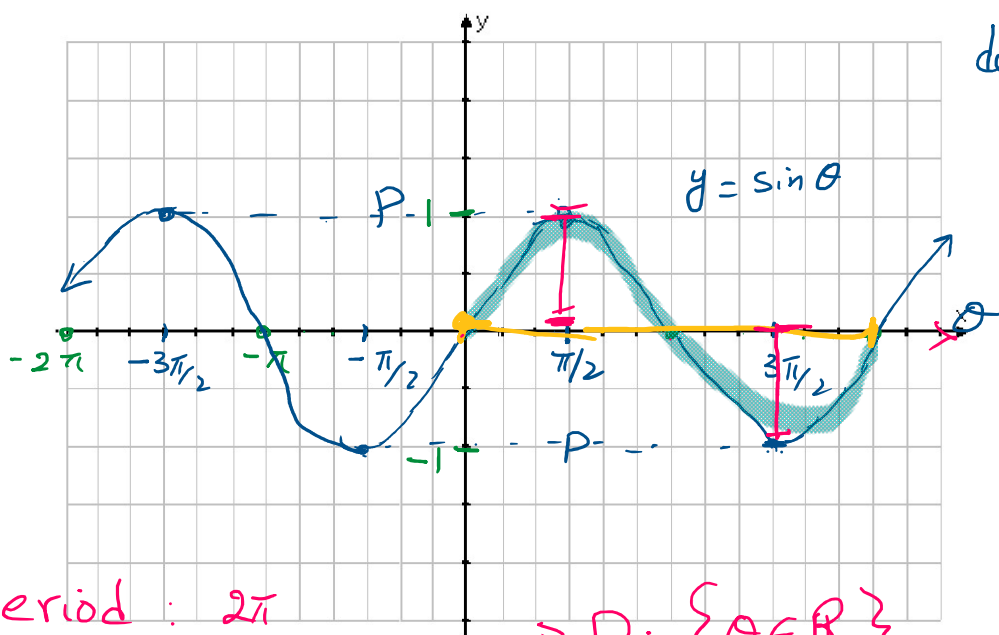


4.1 GRAPHING THE SINE FUNCTIONComplete the table of values for $y = \sin(\theta)$.

θ°	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
Radian Measure	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \sin \theta$	0	1	0	-1	0	1	0	-1	0

Let 6 spaces represent π along the θ -axis. Let 2 spaces along the y -axis represent 1 unit. Scale the axes. Then, plot each (θ, y) point. Then, join the points in a smooth curve. Using your calculator, determine the points in Quadrants II and III. Then, extend your graph to cover the domain $-2\pi \leq \theta \leq 2\pi$.



period: 2π
 zeros @ $n\pi; n \in \mathbb{Z}$
 amplitude: $> 0 (+)$

$$D: \{\theta \in \mathbb{R}\}$$

$$R: \{y \mid -1 \leq y \leq 1, y \in \mathbb{R}\}$$

$$a = \frac{\max - \min}{2}$$

$$= \frac{1 - (-1)}{2}$$

$$= \frac{1+1}{2}$$

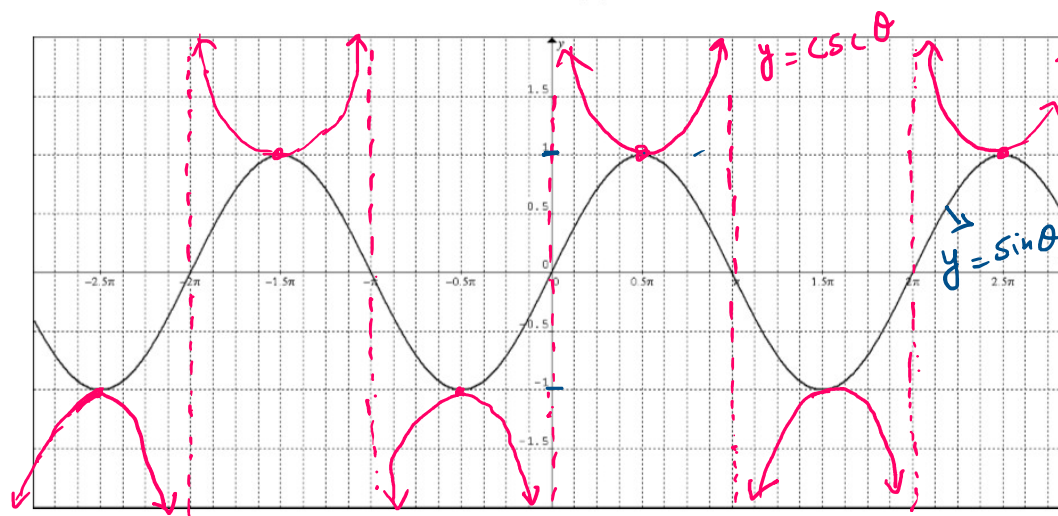
$$a = 1$$

GRAPHING THE COSECANT FUNCTION

Complete the table of values for $y = \csc(\theta)$. Remember, $\csc(\theta) = \frac{1}{\sin(\theta)}$. To find the y-values for the graph $y = \csc \theta$, first evaluate $y = \sin(\theta)$. Then, take the reciprocal. (Your calculator has a $1/x$ button or an x^{-1} button that will give the reciprocal of an input.)

θ°	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
Radian Measure	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \csc \theta = \frac{1}{\sin \theta}$	$\frac{1}{0} = \text{UND}$	1	$\frac{1}{0} = \text{UND}$	-1	$\frac{1}{0} = \text{UND}$	1	UND	-1	UND

Notice if $\sin(\theta) = 0$, then $\csc(\theta)$ is undefined. An undefined value for y implies that there is a vertical asymptote at the corresponding θ value, so draw a vertical dotted line at these locations. Plot the values of $y = \csc(\theta)$ and draw smooth curves through the points for each defined region. Notice that the graph for $y = \sin(\theta)$ has been penciled in so you can make comparisons between it and the graph of $y = \sec(\theta)$.



$$D: \{\theta \in \mathbb{R}, \theta \neq n\pi, n \in \mathbb{Z}\}$$

$$R: \{y \mid y \leq -1, y \geq 1, y \in \mathbb{R}\}$$

$$V.A @ \theta = n\pi, n \in \mathbb{Z}$$

$$(V.A \text{ at all pts } \& \sin \theta = 0)$$

$$y = \frac{1}{x-2}$$

$$V.A @ x=2$$

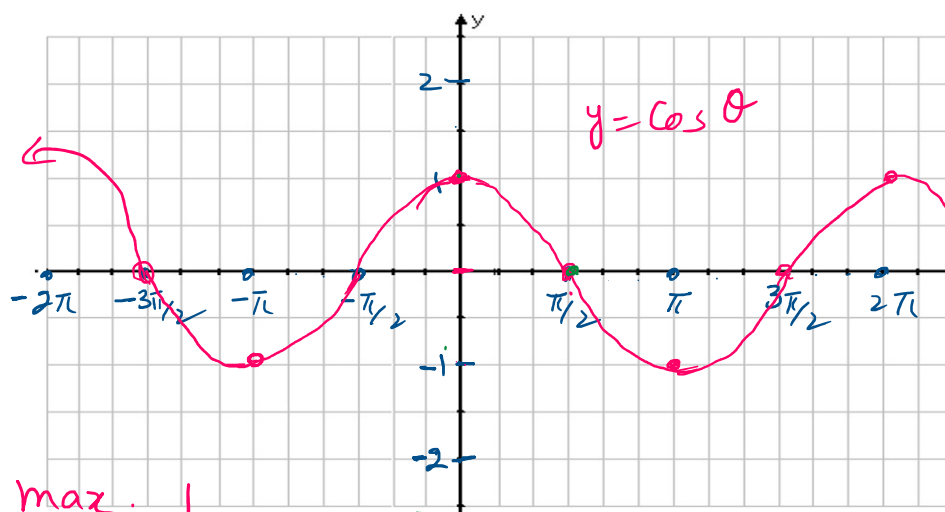
GRAPHING THE COSINE FUNCTION

Complete the table of values for $y = \cos(\theta)$.

θ°	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
Radian Measure	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \cos \theta$	1	0	-1	0	1	0	-1	0	1

$$\cos(-\theta) = \cos \theta$$

Let 6 spaces represent π along the θ -axis. Let 2 spaces along the y -axis represent 1 unit. Scale the axes. Then, plot each (θ, y) point. Then, join the points in a smooth curve. Using your calculator, determine the points in Quadrants II and III. Then, extend your graph to cover the domain $-2\pi \leq \theta \leq 2\pi$.



$$\frac{\pi}{2}$$

Zero's are

$$\textcircled{a} (2n-1)\frac{\pi}{2} \text{ where } n \in \mathbb{Z}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

max: 1
min: -1

amp: 1
 $a > 0 (+)$

period: 2π

$$D: \{\theta \in \mathbb{R}\}$$

$$R: \{y \mid -1 \leq y \leq 1, y \in \mathbb{R}\}$$

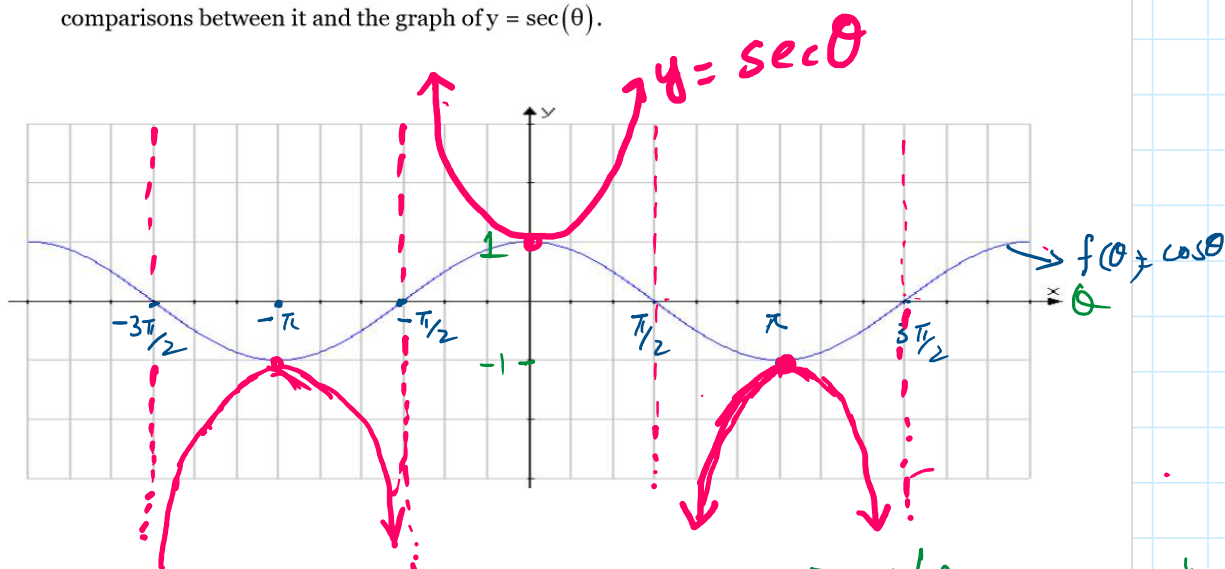
GRAPHING THE SECANT FUNCTION

Complete the table of values for $y = \sec(\theta)$. Remember, $\sec(\theta) = \frac{1}{\cos(\theta)}$. To find the y-

values for the graph $y = \sec(\theta)$, first evaluate $y = \cos(\theta)$. Then, take the reciprocal. (Your calculator has a $1/x$ button or an x^{-1} button that will give the reciprocal of an input.)

θ°	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
Radian Measure	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \sec \theta = \frac{1}{\cos \theta}$		$\frac{1}{0} = \text{UND}$	-1	UND	1	UND	-1	UND	

Notice if $\cos(\theta) = 0$, then $\sec(\theta)$ is undefined. An undefined value for y implies that there is a vertical asymptote at the corresponding θ value, so draw a vertical dotted line at these locations. Plot the values of $y = \sec(\theta)$ and draw smooth curves through the points for each defined region. Notice that the graph for $y = \cos(\theta)$ has been penciled in so you can make comparisons between it and the graph of $y = \sec(\theta)$.



$$D: \{\theta \in \mathbb{R}, \theta \neq (2n-1)\frac{\pi}{2}; n \in \mathbb{Z}\}$$

$$R: \{y \mid y \geq 1, y \leq -1, y \in \mathbb{R}\}$$

V.A are at the points $\cos \theta = 0$

$$\theta = x = (2n-1)\frac{\pi}{2}; n \in \mathbb{Z}$$

$$y = \tan \theta = \frac{\sin \theta}{\cos \theta}$$

GRAPHING THE TANGENT FUNCTION

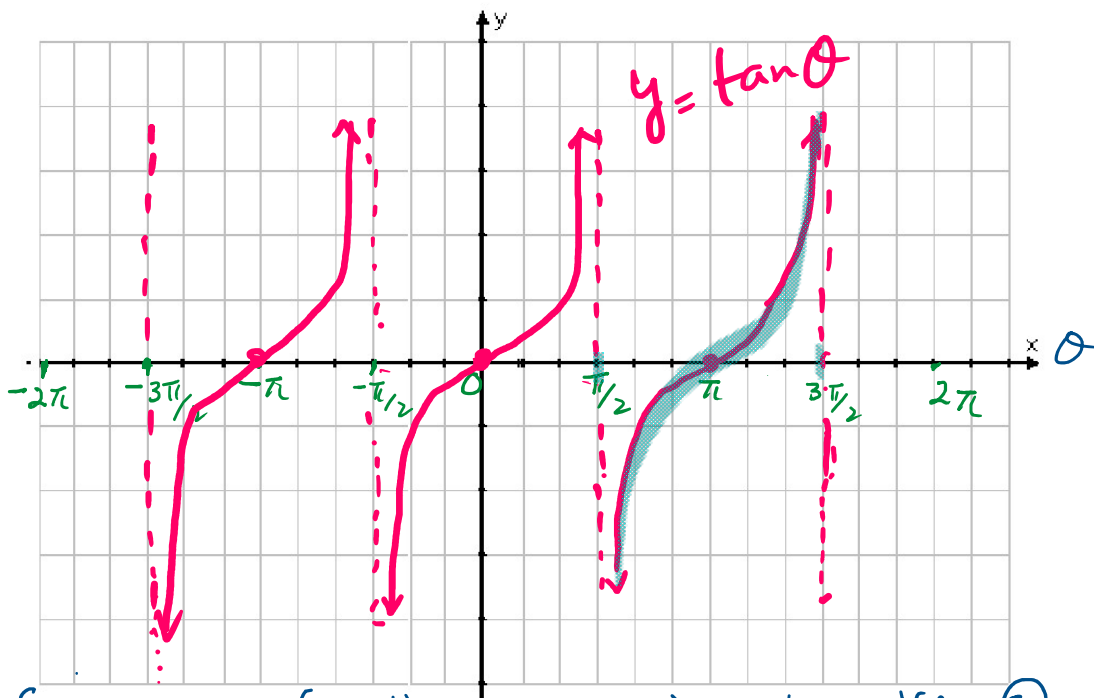
$$\cos \theta = 0 \Rightarrow \text{VA}$$

Complete the table of values for $y = \tan(\theta)$. Then graph the function.

θ°	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
Radian Measure	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \tan \theta$	0	UND	0	UND	0	UND	0	UND	0

Notice if $y = \tan(\theta)$ is undefined, then there is a vertical asymptote at this value for θ .

Draw a dotted line parallel to the y-axis. Your curve should approach these asymptotes closely, but not touch or cross them. Plot the (θ, y) points and draw smooth curves through the points for each defined region.



$$D: \{\theta \in \mathbb{R}, \theta \neq (2n-1)\frac{\pi}{2}, n \in \mathbb{Z}\}$$

$$R: \{y \mid y \in \mathbb{R}\}$$

Period: π

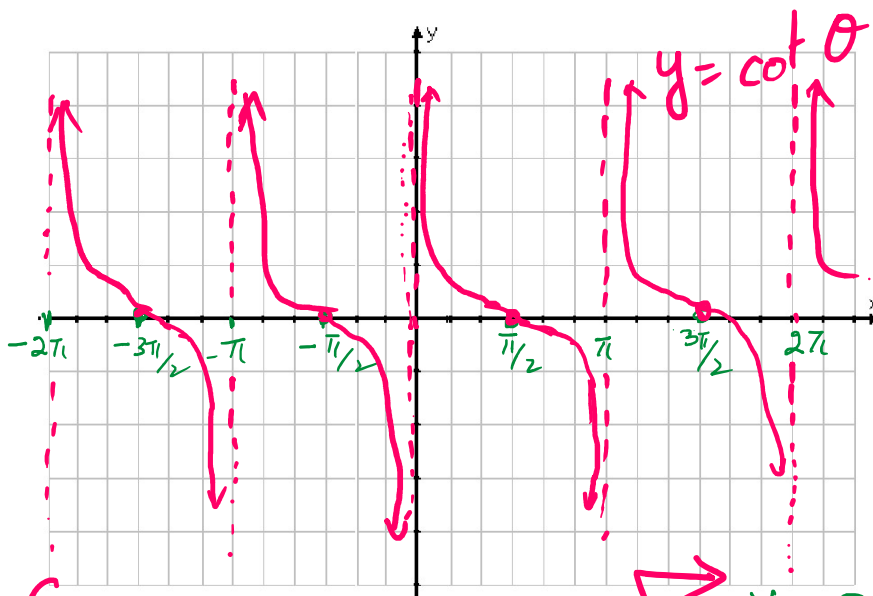
$$\text{VA } \theta = (2n-1)\frac{\pi}{2}, n \in \mathbb{Z}$$

GRAPHING THE COTANGENT FUNCTION

Complete the table of values for $y = \cot(\theta)$. Remember, $\cot(\theta) = \frac{1}{\tan(\theta)}$. To find the y-values for the graph $y = \cot(\theta)$ first evaluate $y = \tan(\theta)$. Then, take the reciprocal. (Your calculator has a $1/x$ button or an x^{-1} button.)

θ°	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
Radian Measure	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \cot \theta$	UND	0	UND	0	UND	0	UND	0	UND

Notice if $\tan(\theta) = 0$, then $\cot(\theta)$ is undefined. An undefined value for y implies that there is a vertical asymptote at the corresponding θ value, so draw a vertical dotted line at these locations. Also, notice that if $\tan(\theta)$ is undefined, then $\cot(\theta) = 0$. Plot the values of $y = \cot(\theta)$ and draw smooth curves through the points for each defined region.



$$D: \{\theta \in \mathbb{R}, \theta \neq n\pi; n \in \mathbb{Z}\}$$

$$R: \{y \mid y \in \mathbb{R}\}$$

$$\text{Period} : \pi$$

$$VA @ x = n\pi; n \in \mathbb{Z}$$

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VA are at the pt where $\tan \theta = 0$ or $\sin \theta = 0$

$$\cot(\theta) = \frac{\cos \theta}{\sin \theta}$$

$\sin \theta$ is zero at $n\pi$ $n \in \mathbb{Z}$